Graph-of-Loops Control for Continuous Autonomous Business Operation

An extension of an already-evaluated department architecture. We ask whether mutually-checking control loops, anchored in code-computed measurement, can sustain a business operated by that architecture without proxy capture.

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Central question

> Can a graph of mutually-checking control loops, anchored in code-computed measurement, hard constraints and human judgement, sustain the continuous autonomous operation of a business without proxy capture — under conditions where a single goal-measure-act loop per department does not?

We do not expect a positive result to show the elimination of proxy capture. The more plausible outcome, if the hypotheses hold at all, is that capture becomes detectable sooner and to a measurable degree.

What is already established

RQ	QUESTION	STATUS	POSITION
RQ1	Multi-agent vs single-LLM baseline	PARTLY MEASURED	Completeness favours decomposition decisively; consistency tied; financial accuracy favours the baseline
RQ2	Dynamic routing efficiency	SUPPORTED	17.3% token saving, 0.0 pp completeness delta
RQ3	Grounding vs parametric knowledge	NOT MEASURABLE	Requires competitor ground truth not yet collected

Two methodological findings are the load-bearing inputs to this proposal. **M1:** cross-agent constraints stated as prompt instructions with explicit numeric bounds were ignored repeatedly, and held only once re-implemented as post-conditions in code. **M2:** variance compounds along a dependency chain — over five runs of an identical prompt the pipeline's standard deviation was 103.8 against the baseline's 21.2, with the pipeline's best run more accurate than the baseline's best.

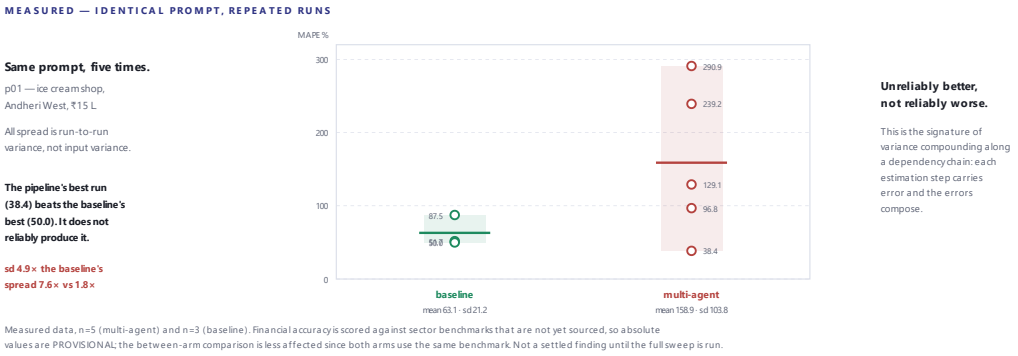


Figure 2 — Five runs of an identical prompt per arm. All spread is run-to-run variance. **Measured**; absolute values provisional pending sourced benchmarks.

Both findings indicate that an agent cannot be relied upon to assert a quantity by which it is judged. We propose to generalise the local remedy — derived figures override asserted ones — into a control architecture for *continuous* operation.

Proposed questions and hypotheses

RQ4 **Does variance compounding persist when the estimated quantity is replaced by one computed from a record of events?**

RQ5 **Does proxy-anchor divergence arise systematically under single-loop control, and how quickly?**

RQ6 **Do independent monitor loops detect divergence earlier than a composite health indicator?**

RQ7 **Are the graph invariants — cycle termination at an anchor or human, monitor independence — necessary for detection, or merely sufficient?**

Six hypotheses are stated in the full proposal, each paired with an explicit falsification condition recorded before the arms are run. Two examples:

H1 A margin computed from a record of transactions will show materially lower run-to-run variance than the same margin estimated through the agent chain.

> **Refuted if:** the computed quantity's standard deviation is not materially below the 103.8 already measured.

H3 Under the full loop-graph topology, terminal proxy-anchor divergence will be materially smaller than under single-loop control.

> **Refuted if:** terminal divergence is statistically indistinguishable from the single-loop arm, or larger.

Method

The existing four-arm harness is extended, not replaced. The current arms — `multi_agent`, `baseline`, `all_departments`, `no_grounding` — remain unmodified, so results already obtained stay valid. Four arms are added: `single_loop`, `anchors_only`, `monitors_only`, `full_graph`. Metrics F54–F58 keep their definitions; F59–F63 are proposed: proxy-anchor divergence, detection latency, conflict incident rate, objective staleness and human decision load.

Every proposed metric is computed by the harness from the append-only record. None derives from any agent's account of its own performance — if the instrument for detecting proxy capture could itself be influenced by the departments under study, the study would exhibit the failure it proposes to measure.

Baseline fairness is the principal methodological risk. The single-loop arm receives the same memory, planning and execution machinery as the full-graph arm, differing only in control topology; its proxy metrics are drawn from measures actually used in comparable business functions rather than chosen for exploitability; and its specification is fixed and documented before any arm runs. We would rather report a null result against a strong baseline than a positive one against a weak baseline.

Pre-committed outcomes

OUTCOME	OBSERVATION	REPORTING
Supporting	Divergence materially lower in the full-graph arm on paired seeds; monitors detect induced capture earlier	As evidence that control topology affects proxy capture, bounded by the simulated environment and scale
Null	Divergence arises but arms are indistinguishable, or divergence does not arise within the horizon	As a null result, with analysis of whether the horizon was short, the perturbations weak, or the construct wrong
Refuting	The full-graph arm diverges equally or more, or its monitors detect later	As a refutation. Oversight adding cost without adding detection is itself a substantive finding

The existing work sets the precedent: its central finding is a negative one, reported rather than tuned away.

Principal threats to validity

- **Construct.** Divergence may conflate Goodhart effects with ordinary action-to-effect lag, anchor mis-specification, or noise — and M2 shows noise here is substantial. The most serious threat, and not fully resolvable.
- **Internal.** Non-determinism may explain between-arm differences; mitigated by paired seeds and distributional reporting. The full topology also performs more inference, so a compute-matched control is needed.
- **External.** The operating environment is simulated; a churn decision is a modelling assumption, not an observation. The prompt set is India-specific.
- **Reflexivity.** The same author designs the architecture, selects the anchors, defines divergence and interprets results. Pre-registration and artefact release are partial mitigations; independent replication would be stronger.

Prerequisites before the proposed arms can be interpreted: the full thirty-five-prompt sweep, sourced sector benchmarks, and collected competitor ground truth. Until then F54 remains provisional and F55 unmeasurable, and any operating-layer result resting on them inherits that status. Full proposal: 14 pp, 6 figures, 16 references.